

Probability Measures

Math Foundations – Node 19

1 Setting

Throughout, Ω is a non-empty set (the *sample space*) and $\mathcal{F} \subseteq 2^\Omega$ is a σ -algebra on Ω in the sense of node 18: $\Omega \in \mathcal{F}$, $\mathcal{F}$ is closed under complements, and $\mathcal{F}$ is closed under countable unions. Elements of $\mathcal{F}$ are called *events*.

2 Definitions

Definition 1 (Probability measure). A *probability measure* on $(\Omega, \mathcal{F})$ is a set function $P : \mathcal{F} \rightarrow [0, 1]$ satisfying the three *Kolmogorov axioms*:

(K1) *Non-negativity*: $P(A) \geq 0$ for every $A \in \mathcal{F}$.

(K2) *Normalisation*: $P(\Omega) = 1$.

(K3) *Countable additivity* (σ -additivity): for every sequence $\{A_n\}_{n \geq 1} \subseteq \mathcal{F}$ of pairwise disjoint events,

$$P\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} P(A_n).$$

Definition 2 (Probability space). A *probability space* is a triple $(\Omega, \mathcal{F}, P)$ where Ω is a sample space, $\mathcal{F}$ is a σ -algebra on Ω , and P is a probability measure on $(\Omega, \mathcal{F})$.

Remark 3. From (K1)–(K3) one immediately obtains $P(\emptyset) = 0$ (take all $A_n = \emptyset$ in (K3); the sum $\sum P(\emptyset)$ converges only if each term vanishes) and finite additivity $P(A \sqcup B) = P(A) + P(B)$ by padding with empty sets.

3 The inclusion–exclusion identity

Theorem 4 (Inclusion–exclusion, two-set form). *Let $(\Omega, \mathcal{F}, P)$ be a probability space. For every $A, B \in \mathcal{F}$,*

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

Proof. We decompose $A \cup B$ and B into disjoint pieces and apply finite additivity, which follows from (K3) as noted above.

Step 1: $A \cup B = A \sqcup (B \setminus A)$ **disjointly**. Every $\omega \in A \cup B$ is in A or in $B \setminus A := B \cap A^c$, and these two sets are disjoint by construction. Both lie in $\mathcal{F}$: $A \in \mathcal{F}$ by hypothesis, and $B \setminus A = B \cap A^c \in \mathcal{F}$ since $\mathcal{F}$ is closed under complements and finite intersections. By finite additivity,

$$P(A \cup B) = P(A) + P(B \setminus A). \tag{1}$$

Step 2: $B = (B \cap A) \sqcup (B \setminus A)$ **disjointly**. Every $\omega \in B$ either lies in A (hence in $B \cap A$) or does not (hence in $B \setminus A$), and these two sets are disjoint. Both lie in $\mathcal{F}$ for the same reasons as in Step 1. By finite additivity,

$$P(B) = P(B \cap A) + P(B \setminus A),$$

and rearranging gives

$$P(B \setminus A) = P(B) - P(A \cap B). \quad (2)$$

(Note $P(B \cap A) = P(A \cap B)$ since intersection is commutative.)

Step 3: combine. Substituting (??) into (??),

$$P(A \cup B) = P(A) + (P(B) - P(A \cap B)) = P(A) + P(B) - P(A \cap B). \quad \square$$

Remark 5. The general n -set inclusion–exclusion formula

$$P\left(\bigcup_{i=1}^n A_i\right) = \sum_{k=1}^n (-1)^{k+1} \sum_{1 \leq i_1 < \dots < i_k \leq n} P(A_{i_1} \cap \dots \cap A_{i_k})$$

follows by induction on n , with Theorem ?? as the base case.

4 An immediate corollary

Proposition 6 (Boole’s inequality). *For every countable collection $\{A_n\}_{n \geq 1} \subseteq \mathcal{F}$,*

$$P\left(\bigcup_{n=1}^{\infty} A_n\right) \leq \sum_{n=1}^{\infty} P(A_n).$$

Proof. Set $B_1 = A_1$ and $B_n = A_n \setminus \bigcup_{k < n} A_k$ for $n \geq 2$. The B_n lie in $\mathcal{F}$, are pairwise disjoint, and satisfy $\bigcup_n B_n = \bigcup_n A_n$ with $B_n \subseteq A_n$. By (K3) and monotonicity (itself a one-line consequence of Theorem ?? applied with $B \supseteq A$),

$$P\left(\bigcup_n A_n\right) = \sum_n P(B_n) \leq \sum_n P(A_n). \quad \square$$